

Mitchell B Slapik

MD/PhD Candidate & NIH F30 Fellow, McGovern Medical School, mslapik@gmail.com

Education

2027	McGovern Medical School, Houston, TX Medical Scientist Training Program	MD
2025	Graduate School of Biomedical Sciences <i>Houston, TX, Neuroscience</i>	PhD
2017	Johns Hopkins University, Baltimore, MD Post-Baccalaureate Premedical Program	
2014	Swarthmore College, Swarthmore, PA With High Honors in Philosophy and Linguistics	BA
2013	University of Oxford, Oxford, UK Study Abroad: Philosophy of Mind	

Publications

2026	R. Milton*, M. Slapik* , S. Egranov, et al. "Increased Visuo-Frontal Communication Enhances Environmental Representation During Locomotion." Accepted in principle at <i>Nature Neuroscience</i> . * <i>Equal contribution</i> . M. Slapik , H. Shouval. "Simulated Complex Cells Contribute to Object Recognition through Representational Untangling." <i>Neural Computation</i> .
2025	M. Slapik . "Computational Evidence for an Inverse Relationship between Brain and Retinal Complexity." <i>Journal of Vision</i> .
2022	M. Joyce, P. Nadkarni, S. Kronemer, [...], M. Slapik , et al. "Quality of Life Changes Following the Onset of Cerebellar Ataxia: Symptoms and Concerns Self-reported by Ataxia Patients and Informants." <i>Cerebellum</i> .
2020	O. Morgan, M. Slapik , K. Iannuzzelli, et al. "The Cerebellum and Sequencing in Motor and Cognitive Domains: Evidence from Cerebellar Ataxia." <i>Cerebellum</i> . S. Kronemer, M. Slapik , J. Pietrowski, et al. "Neuropsychiatric Symptoms as a Reliable Phenomenology of Cerebellar Ataxia." <i>Cerebellum</i> .

2018 **M. Slapik**, S. I. Kronemer, O. Morgan, et al. "Visuospatial Organization and Recall in Cerebellar Ataxia." *Cerebellum*.

Manuscripts in preparation

M. Slapik, et al. "Intermediate Representations Reveal Reproducible Structure Beyond Vision Foundation Models in Human V4."

M. Slapik. "Adaptive Systems Selectively Regulate Access to Their Own Internal States."

X. Niu, **M. Slapik**, A. McConnell, et al. "Prefrontal Population Activity Tracks Continuous Whole-Body Configuration During Natural Behavior."

J. Kim, M. Franch, **M. Slapik**, et al. "Social Learning Refines dIPFC Context Encoding by Selectively Recruiting Context-Informative Neurons."

Research experience

2021 – Now **MD/PhD Graduate Researcher, Dragoi Lab**
McGovern Medical School, *Houston, TX*

Investigate neural population dynamics, inter-areal communication, and biological and artificial representations of natural behavior and vision

- Analyzed neural activity in V4 and dIPFC during free exploration, and quantified inter-areal communication using reduced-rank regressions and spike-timing coordination
- Studied how early visual computations contribute to representational untangling in biological and artificial networks.
- Investigated the relationship between retinal and brain complexity using convolutional neural networks
- Designed optimal stimuli for neurons in visual cortex using an image generator and optimizer

2016 - 2019 **Research Assistant, Marvel Lab**
Johns Hopkins School of Medicine, *Baltimore, MD*

Examined the cognitive and emotional effects of cerebellar ataxia

- Designed new cognitive tasks assessing visuospatial skills, gestalt processing, sequence learning and verbal encoding
- Administered cognitive tasks, emotional questionnaires and motor tests to ataxia patients and controls

- Analyzed behavioral and questionnaire data, contributed to peer-reviewed manuscripts and presented findings to researchers, clinicians and patient communities

Fellowships and awards

2023	NIH Ruth L. Kirschstein NRSA Individual Predoctoral Fellowship (F30) National Eye Institute 1F30EY035603-01 “Optimal Stimuli as a New Method to Investigate Neural Networks”
2022	Clinical and Translational Science Predoctoral Fellowship (TL1) National Institutes of Health: National Center for Advancing Translational Sciences
2024	Dean’s Research Scholarship McGovern Medical School
2023	George M. Stancel Fellowship in the Biomedical Sciences McGovern Medical School
2023	Best Work-In-Progress Talk Department of Neurobiology and Anatomy, McGovern Medical School
2022, 2023, 2024	Osborne Endowed Scholarship in the Neurosciences Department of Neurobiology and Anatomy, McGovern Medical School
2018	Travel Award for 8 th Annual Ataxia Investigator’s Meeting The National Ataxia Foundation
2014	High Honors in Philosophy and Linguistics Swarthmore College Honors Program

Presentations

Selected talks

2024	M. Slapik. “Retinal Complexity Varies Inversely with Brain Complexity in a Computational Model.” Post-candidacy Talk at Fall Neuroscience Retreat for UTHealth. Houston, TX.
2023	M. Slapik, H. Shouval. “Unshattering Dimensionality.” Post-candidacy Talk at Fall Neuroscience Retreat for UTHealth. Cleveland, TX.

- 2022 **M. Slapik**, A. Andrei, S. Khan, et al. "Optimal Stimuli as a New Method to Investigate Neural Networks." Pre-candidacy Talk at Fall Neuroscience Retreat for UTHealth. Cleveland, TX.
- M. Slapik**, O. Morgan, J. Creighton, et al. "Timing and Sequencing in Cerebellar Ataxia." Nanosymposium talk accepted for presentation at: Society for Neuroscience San Diego, CA.
- M. Slapik**, O. Morgan, C. Marvel. "Language Abilities in Cerebellar Ataxia." Presentation to the faculty and staff of the Johns Hopkins Ataxia Clinic, Baltimore, MD.
- 2017 **M. Slapik**, S. Kronemer, O. Morgan, et al. "Visuospatial Organization and Recall in Cerebellar Ataxia." Talk presented at: Sensorimotor Day, Johns Hopkins University, Baltimore, MD.

Selected posters

- 2024 **M. Slapik**, X. Niu, A. McConnell, et al. "Executive Regions Encode Movement of Individual Body Parts in Macaque Monkeys." Society for Neuroscience. Chicago, IL.
- M. Slapik**, H. Shouval. "The Visual System Relies on Organized but Low-Dimensional Representations for Object Recognition." Spring Neuroscience Retreat for UTHealth, Houston, TX.
- 2023 **M. Slapik**, A. Andrei, S. Khan, et al. "A Deep Learning Approach to Naturalistic Surround Modulation." Society for Neuroscience. Washington, D.C.
- 2022 **M. Slapik**, A. Andrei, S. Khan, et al. "Deep Networks Design Optimal Stimuli for Early Visual Cortex." Society for Neuroscience. San Diego, CA.
- 2020 **M. Slapik**, S. Patwardhan, R. Costa, et al. "Using Machine Learning To Classify Feeding Behavior in Aplysia." American Physician Scientists Association. Houston, TX.
- 2018 **M. Slapik**, J. Pietrowski, O. P. Morgan, et al. "A Characterization of Language Impairment in Cerebellar Ataxia." The National Ataxia Foundation's 8th Ataxia Investigator's Meeting. Philadelphia, PA.
- 2017 **M. Slapik**, S. Kronemer, J. Mandel, et al. "Visuospatial Processing and Strategy Formation in Cerebellar Ataxia." Society for Neuroscience, Washington, D.C.

Clinical and community service

- 2026 - Now **Curriculum Developer**, AI Standardized Patient Project
McGovern Medical School, *Houston, TX*
- Developed an AI-based standardized patient simulation for third-year medical students to practice psychiatric interviewing, diagnostic assessment, and clinical reasoning.
 - Collaborated with psychiatry faculty to design case scenarios, learning objectives, and structured feedback for integration into the MS3 psychiatry curriculum.
- 2021 – Now **Volunteer Counselor**
Crisis Text Line, *Houston, TX*
- Provide crisis support via text to individuals experiencing suicidal ideation, self-harm urges, anxiety, depression, and acute psychosocial stressors
 - Connect texters to personalized resources related to depression, anxiety, substance use, housing insecurity, and gender/sexual identity
- 2017 – 2019 **Team Leader, Health Resource Coordinator**
Charm City Clinic, *Baltimore, MD*
- Assisted clients with a variety of challenges related to medical care, employment, and housing
 - Trained a new group of volunteers
- 2015 – 2016 **Emergency Room Volunteer**
Penn Presbyterian Medical Center, *Philadelphia, PA*
- Took incoming calls, paged nurses, restocked supplies, and observed procedures
- 2015 – 2016 **Front-Desk Volunteer**
Washington West Project, *Philadelphia, PA*
- Enrolled patients for STD screening and counseling

Research service

Mentoring

Xiaoke Niu, Jaehwan Kim, Joey Zambelas, Madeline Gomez

Reviewing

Scientific Reports, Frontiers in Computational Neuroscience, Cerebellum

Machine Learning Journal Club

Leadership Committee

Organizations

2026 – Now **American Psychiatric Association**

2021 – Now **American Physician Scientists Association**

2019 – Now **American Medical Association**

2017 – Now **Society for Neuroscience**

2017 – 2019 **National Ataxia Foundation**

Skills

Electrophysiology: Acute and chronic recording, spike-sorting, eye tracking, receptive field mapping, behavioral task design

Neuroimaging: fMRI analysis, encoding models, representational analysis

Data analysis: MATLAB, Python, SPSS

Machine Learning: PyTorch, scikit-learn, vision foundation models, dimensionality reduction, decoding, regression

Task Development: E-Prime, PsychToolbox, MonkeyLogic

Languages

English: Fluent

Spanish: Basic

Interests

Jazz Saxophone
The Chirp Chirps, Bayou City Funk

Classical and Jazz Piano